

KARTHIK M

📍 No. 21, Mathru Layout, Near Dairy Circle, Yelahanka New Town, Bengaluru - 560064, Karnataka, India

Phone : [9741166419](tel:9741166419) • Email : karthikraj1996@gmail.com • LinkedIn : [karthik-m](#)

Software Developer with 7+ years of experience in building robust desktop applications using C++, C#, and SQL. Specialized in developing scalable healthcare and embedded solutions with a focus on performance, code quality, and Agile delivery.

Skills

Programming Languages: C++ (C++11/14/17), C, C#, SQL

Tools & IDEs: Visual Studio 2013/2019/2022, SSMS, Installshield, Wix Toolset, Windbg, Process Explorer, SonarQube, ReSharper,

Unit Testing: MSTest Unittest framework, Hippomocks

Version Control & Collaboration: Git (Git CLI, GitHub, Bitbucket), Jira / Azure DevOps / Confluence

Operating System: Windows, macOS

Experience

Software Technologist I

Philips

Bengaluru, India

Jan 2022 – Present

- Actively contributed to the design, development, and continuous enhancement of the Philips EPIQ/Affiniti product line, which provides state-of-the-art ultrasound imaging solutions. These systems are equipped with clinically optimized tools specifically designed to improve diagnostic accuracy, enhance productivity, and support informed clinical decision-making across a wide range of medical domains.
- Played a key role in the implementation of multiple product features and the resolution of numerous software issues across several major platform releases, including VM10, VM11, VM12, and VM14. These efforts involved close collaboration with cross-functional teams to ensure robust, high-quality software delivery within the framework of Agile development cycles.
- Developed core application logic using C++ and C#, with a strong focus on performance and reliability. Leveraged multithreading techniques to handle parallel processing requirements, improving the responsiveness and throughput of critical modules within the system.
- Investigated and resolved complex multithreading issues, including deadlocks and race conditions, by analyzing thread interactions and synchronization mechanisms. Implemented efficient solutions to eliminate concurrency flaws, significantly enhancing application stability and performance under high-load conditions.
- Served as the Feature Lead for the VM12.0 Annotation Improvements, a significant enhancement feature aimed at increasing the efficiency and usability of the annotation functionality within the ultrasound system. This effort included multiple workflow refinements and UI/UX improvements designed to streamline diagnostic workflows for clinicians.
- Currently engaged in the design and development of the VM14.0 Koios Integration feature, which incorporates AI-powered diagnostic support into the platform (uses REST API to communicate to server). This feature aims to assist clinicians in the early detection and evaluation of breast and thyroid cancer, contributing to improved diagnostic outcomes.
- Delivered key contributions within the Analysis Domain, particularly in the development of functionalities for the calculation, aggregation, and visual presentation of diagnostic results on the user interface. This has significantly enhanced the clarity, accessibility, and interpretability of clinical data for end users.
- Developed and maintained unit tests using the MSTest framework, and employed HippoMocks to create isolated test environments by mocking dependencies. This ensured code reliability, facilitated early defect detection, and supported long-term maintainability.
- Wrote efficient SQL queries and developed stored procedures to retrieve, store, and manipulate data within

the system's backend, ensuring data consistency and optimized performance across various modules.

Senior Software Engineer

Mindtree

Bengaluru, India

Oct 2018 – Dec 2021

PROJECT : HSUPRO

- HSUPro is a native Windows-based desktop application designed for Sennheiser device users to manage firmware updates, device configurations, and asset information, tailored for enterprise use in embedded desktop environments.
- Developed new features and integrated support for new devices, contributing to the extensibility and adaptability of the platform.
- Troubleshoot and resolved critical issues, demonstrating strong problem-solving skills and attention to detail.
- Involved throughout the entire software development life cycle, from requirement analysis to deployment and maintenance.
- Followed Agile software development practices, including participation in sprint planning, code reviews, and continuous integration.
- Created and managed InstallShield-based installer packages to support seamless deployment and upgrades.
- Independently managed and supported the application's development for a period of over six months, showcasing initiative, reliability, and end-to-end ownership.

PROJECT : EPOS CONNECT

- EPOS Connect is a user-friendly, cross-platform desktop application(Windows, Thin Client, Mac) designed to support asset management, firmware updates, and configuration management for EPOS audio devices. It ensures seamless connectivity and lifecycle management for users across multiple platforms.
- Contributed extensively to a shared codebase reused across diverse platforms including Windows, macOS, and thin clients, enabling modular development and efficient maintenance.
- Implemented new features and device support using modern C++11, adhering to clean coding practices and ensuring cross-platform compatibility.
- Resolved complex bugs through in-depth debugging techniques, utilizing log analysis, static analysis tools, and platform-specific troubleshooting methods.
- Applied STL, multithreading, and advanced C++ paradigms to develop efficient and high-performing code.
- Ensured high code quality through static code analysis, maintaining a clean and maintainable codebase free from code smells.
- Actively participated in Agile development processes, including sprint planning, task estimation, and ownership of development stories, contributing to consistent and timely delivery of new features.

Education

SJB Institute of Technology

Bachelor of Engineering in Computer Science and Engineering; 72.08%

Bengaluru, India

June 2018

Leaders' PU College

Pre-University Course(10+2); 87.17%

Gauribidanur, India

July 2014

Vidyanidhi Public School

Class 10 (ICSE); 82.82%

Gauribidanur, India

April 2012

Certifications

- Object Oriented Data Structures in C++, *Coursera*. 2020
- C++ Programming, *Aprameyah Technologies*. 2019
- C++ STL Programming, *Aprameyah Technologies*. 2020
- Identifying Security Vulnerabilities in C/C++ , *Coursera*. 2021

Awards & Achievements

- Received SPOT award from Philips (An award given by Philips to its employees for working on critical deliverables).
- Recognized by Manager with A-Team award 5 times (An award given by Mindtree to its employees for their excellent team effort).
- Awarded with Unstoppable and HatsOff Awards (An award given by Mindtree to its employees for their excellent contribution).